Use the same dataset you created in the previous assignment and implement the following.

1. Implement two kinds of normalization to compute the TF
    a. Normalization based on the total number of words in the sentence
    b. Normalization based on the frequency of the most frequent word in the sentence
2. Implement the normalization of IDF based on the most frequent word across sentences
3. Design the logistic regression based on the normalized TF and IDF vectors and compare the performance with the unnormalized TF and IDF vectors. Try out all combinations of normalized and unnormalized TF and IDF vectors. You will have total 6 types of combinations. You have already developed a model for the unnormalized TFs and IDFs in the previous assignment.